University of Florida, Department of Mathematics

Partial Differential Equations PhD Exam, May 2010

1. Suppose $u \in C_1^2(\mathbb{R}^n \times [0,T]) \cap C(\mathbb{R}^n \times [0,T])$ solves

$$\begin{cases} u_t - \sum_{i,j=1}^{n} a_{ij}\partial_{ij} u = 0, & \text{in } \mathbb{R}^n \times (0,T], \\ u = g, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

where $(a_{ij})_{n\times n}$ is a positive definite, symmetric constant matrix, g is a given smooth function. Suppose in addition that

$$u(x,t) \le Ae^{a|x|^2}, \qquad x \in \mathbb{R}^n, \quad t \in [0,T]$$

for some $a, A > 0$. Prove

$$\sup_{\mathbb{R}^n \times [0,T]} u = \sup_{\mathbb{R}^n} g.$$

2. Let u solve

$$\begin{cases} u_{tt} - u_{xx} = 0, & \mathbb{R}_+ \times (0,\infty), \\ u = g, \quad u_t = h, & \text{on } \mathbb{R}_+ \times \{t = 0\}, \\ u = 0, & \text{on } \{x = 0\} \times (0,\infty). \end{cases}$$

where g, h are smooth, given functions of x such that $g(0) = h(0) = 0$. Write the explicit expression of u.

3. Let B_1 be the unit ball in $\mathbb{R}^n$ $(n \ge 3)$, $p \in \left(1, \frac{2n}{n-2}\right)$ and $\alpha \in (0,1)$, show that there exists a C depending only on n, p, α such that

$$\left(\int_{B_1} |u|^p \, dx\right)^{1/p} \le C(n,p,\alpha) \left(\int_{B_1} |\nabla u|^2\right)^{1/2},$$

for all $u \in H^1(B_1)$ that satisfy the following property:

$$|\{x \in B_1;\ u(x) = 0\}| \ge \alpha |B_1|.$$

4. Suppose $u \in C^2$ solves

$$u_{tt} - \Delta u = 0 \qquad \text{in } \mathbb{R}^n \times (0,\infty).$$

Fix $x_0 \in \mathbb{R}^n$, $t_0 > 0$ and consider the cone

$$C = \{(x,t);\ 0 \le t \le t_0,\ |x - x_0| \le t_0 - t\}.$$

Prove that if $u \equiv u_t \equiv 0$ on $B(x_0, t_0) \times \{t = 0\}$, then $u \equiv 0$ in C.

5. Let $(a_{ij}(x))_{n\times n}$ be smooth, symmetric on B_1 and there exist $0 < \lambda < \Lambda$ such that

$$\lambda |\xi|^2 \le \sum_{i,j=1}^{n} a_{ij}(x)\xi_i\xi_j \le \Lambda |\xi|^2, \qquad \forall x \in B_1, \quad \forall \xi \in \mathbb{R}^n.$$

Let b_i $(i = 1, 2, \dots, n)$, c and f be smooth functions on B_1. Show that there exists $\delta > 0$ such that if

$$\|b_i\|_{L^\infty(B_1)} + \|c\|_{L^\infty(B_1)} \le \delta, \qquad i = 1, \dots, n,$$

then there exists a smooth function u that solves the Dirichlet boundary problem uniquely:

$$\begin{cases} -\sum_{i,j=1}^{n} \partial_i\big(a_{ij}(x)\partial_j u\big) + \sum_{i=1}^{n} b_i(x)\partial_i u(x) + c(x)u(x) = f, & B_1, \\ u = 0, & \text{on } \partial B_1. \end{cases}$$

6. Let $\{u_m\}$ be a sequence of smooth functions satisfying $u_m(x) \leq C$ and $|\Delta u_m| \leq C$ for some constant C uniformly over B_1, the unit ball. Prove that over $B_{1/2}$, along a subsequence, either u_m converges to $-\infty$ uniformly, or u_m converges to a function u in C^2 norm.

7. Warning: Suspected error. The displayed variational form places the lower-order terms inside the scope of the outer sum over i, j, while also using an inner sum over i. This potentially unintended summation scope has been retained because the intended correction is not uniquely determined by the immediate context.

Let $\{a_{ij}\}_{n\times n}$ $(n \geq 2)$ be a positive definite, symmetric, constant matrix, $b_1, \dots, b_n, c$ be smooth functions on B_1, the unit ball in $\mathbb{R}^n$. Show that the minimum of the following variational form

$$I(w) = \int_{B_1} \sum_{i,j=1}^{n} \left(a_{ij}\partial_i w \partial_j w + \sum_i b_i \partial_i w\, w + cw^2 \right) dx, \qquad w \in H_0^1(B_1)$$

under the constraint: $\int_{B_1} |w|^2 = 1$ can be reached by a smooth function.